



CSCE 633-700: Machine Learning

Clustering

Overview

- Clustering overview
- Partitional clustering
 - K-means clustering
 - Gaussian Mixture Models (GMM)
- Hierarchical clustering

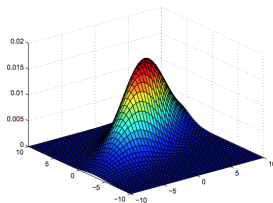
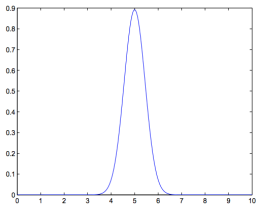
Multivariate Gaussian distribution

Univariate Gaussian distribution

$$p(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

Multivariate Gaussian distribution

$$p(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$



Multivariate Gaussian distribution

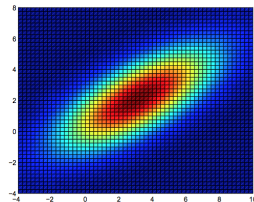
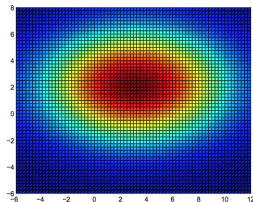
Covariance matrix

- Covariance between two random variables X and Y
$$\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y))) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$
- The covariance matrix provides a way to summarize the covariances of all pairs of variables $(\Sigma)_{ij} = \text{Cov}(X_i, X_j)$
- Σ is always positive definite

Multivariate Gaussian distribution

Isocontours

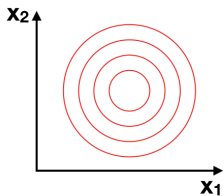
- For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ an isocontour is a set of the form $\{\mathbf{x} \in \mathbb{R}^2 : f(\mathbf{x}) = c\}$



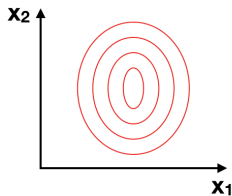
Multivariate Gaussian distribution

The diagonal covariance matrix

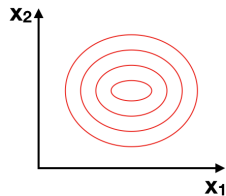
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$



$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



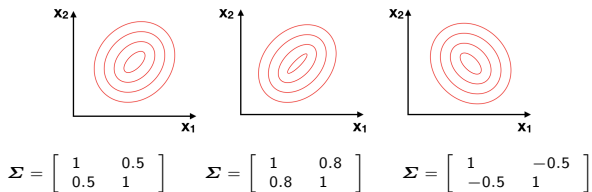
$$\boldsymbol{\Sigma} = \begin{bmatrix} 0.6 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Multivariate Gaussian distribution

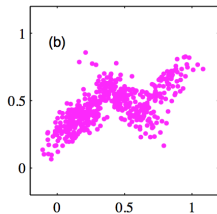
The non-diagonal covariance matrix



By increasing the off-diagonal elements from 0.5 to 0.8, the distribution is more thinly peaked along the line where x_1 is equal to x_2

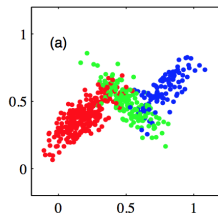
Probabilistic interpretation of clustering

- We want to find $p(\mathbf{x})$ that best describes our data
- The data points seem to form 3 clusters
- However, we cannot model $p(\mathbf{x})$ with simple and known distributions, e.g. one Gaussian



Probabilistic interpretation of clustering

- Instead, we will model each region with a Gaussian distribution → Gaussian mixture models (GMMs)
- Question 1: How do we know which (color) region a data point comes from?
- Question 2: What are the parameters of Gaussian distributions in each region?
- We will answer both in an unsupervised way from data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$



GMM as the marginal distribution of a joint distribution

- The joint distribution between $\mathbf{x}$ and z (representing color) are

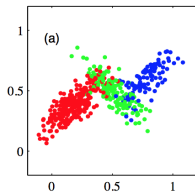
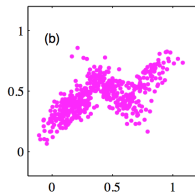
$$p(\mathbf{x}|z = red) = \mathcal{N}(\mathbf{x}; \mu_1, \Sigma_1)$$

$$p(\mathbf{x}|z = blue) = \mathcal{N}(\mathbf{x}; \mu_2, \Sigma_2)$$

$$p(\mathbf{x}|z = green) = \mathcal{N}(\mathbf{x}; \mu_3, \Sigma_3)$$

- The marginal distribution is thus

$$p(\mathbf{x}) = p(red)\mathcal{N}(\mathbf{x}; \mu_1, \Sigma_1) + p(blue)\mathcal{N}(\mathbf{x}; \mu_2, \Sigma_2) \\ + p(green)\mathcal{N}(\mathbf{x}; \mu_3, \Sigma_3)$$



Gaussian mixture models

A Gaussian mixture model has the following density function for $\mathbf{x}$

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

- K : number of Gaussians
- $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$: mean & covariance of k^{th} component
- π_k : component weights

$$\pi_k > 0, \quad \forall k \quad \text{and} \quad \sum_{k=1}^K \pi_k = 1$$

- Estimate $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \pi_k \Rightarrow$ Expectation Maximization

Parameter estimation for GMMs

If we know the probability of sample $\mathbf{x}_n$ belonging to Gaussian component k , i.e., responsibility $\gamma(z_{nk})$, we can estimate the parameters of each Gaussian distribution $\{\mu_k, \Sigma_k, \pi_k\}$
(Maximization Step)

$$\pi_k = \frac{\sum_n \gamma(z_{nk})}{\sum_k \sum_n \gamma(z_{nk})} \quad \mu_k = \frac{1}{\sum_n \gamma(z_{nk})} \sum_n \gamma(z_{nk}) \mathbf{x}_n$$

$$\Sigma_k = \frac{1}{\sum_n \gamma(z_{nk})} \sum_n \gamma(z_{nk}) (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

- For π_k : count the number of data points whose z_n is k and divide by the total number of data points
- For μ_k : the mean of all samples weighted by their responsibility (i.e., probability of belonging to mixture k)
- For Σ_k : the covariance matrix of all samples weighted by their responsibility (i.e., probability of belonging to mixture k)

Parameter estimation for GMMs: incomplete data

If we know the parameters of each Gaussian mixture $\{\mu_k, \Sigma_k, \pi_k\}$, we can find the probability of each data sample $\mathbf{x}_n$ belonging to Gaussian mixture k (Expectation Step)

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \mu_j, \Sigma_j)}$$

Every data point $\mathbf{x}_n$ is assigned to a component fractionally according to $\gamma(z_{nk})$, also called **responsibility**

Parameter estimation for GMMs

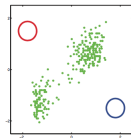
Since we do not know μ_k , Σ_k to begin with, we cannot compute $\gamma(z_{nk})$ or π_k

But we can invoke an iterative procedure and alternate between estimating γ_{nk} using π_k , μ_k and Σ_k , and vice-versa.

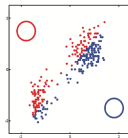
- **Step 0:** Guess π_k , μ_k , Σ_k with initial values
- **Step 1 (E-Step):** Compute γ_{nk} using current π_k , μ_k , Σ_k
- **Step 2 (M-Step):** Update π_k , μ_k , Σ_k using computed γ_{nk}
- **Step 3:** Go back to Step 1

Parameter estimation for GMMs

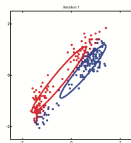
Example of GMM parameter estimation with EM



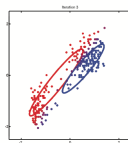
(a)



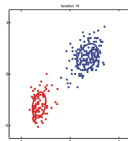
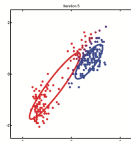
(b)



(c)

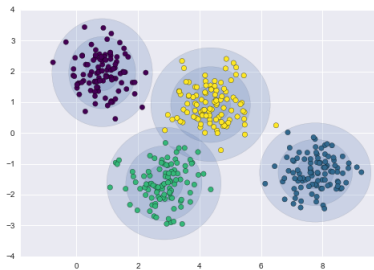


(d)



Gaussian Mixture Models

Example of GMM parameter estimation with EM



Gaussian Mixture Models

Comparison between K-Means and GMMs

